

BRANDON M. DURAN

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RESEARCH INTERESTS

aerosol-cloud interactions, aerosol radiative forcing, circulation responses to aerosol forcing, satellite simulators

EDUCATION

Scripps Institution of Oceanography, University of California San Diego 2022 - Present
Masters in Earth Science (2024), Ph.D. expected in 2027
Advisors: Profs. Nicholas Lutsko & Shang-Ping Xie
Committee Members: Profs. Duncan Watson-Parris, Casey Wall, & Amato Evan
Georgetown University 2018-2022
BA in Physics, Computer Sciences

PEER REVIEWED PUBLICATIONS

4. **Duran, B.M.**, Lutsko, N.J., and Wall, C.J. Aerosol-Ice-Cloud Interactions in a Perturbed Parameter Ensemble (2026). *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-25-0724.1>.
3. Michibata, T., Wall, C.J., Hirota, N., **Duran, B.M.**, and Nozawa, T. Recent Advances in the Observation and Modeling of Aerosol-Cloud Interactions, Cloud Feedbacks, and Earth's Energy Imbalance: A Review (2025). *Current Pollution Reports*. <https://doi.org/10.1007/s40726-025-00382-6>.
2. Wall, C.J., Paynter, D., Qin, Y., Debolskiy, M., Duffy, M.L., Michibata, T., **Duran, B.M.**, Lutsko, N.J., Ma, P.L., Medeiros, B., Storelvmo, T., and Zhao, M. Decomposing Cloud Radiative Feedbacks by Cloud-Top Phase (2025). *Journal of Climate*. <https://doi.org/10.1175/JCLI-D-24-0538.1>.
1. **Duran, B.M.**, Wall, C.J., Lutsko, N.J., Michibata, T., Ma, P.L., Qin, Y., Duffy, M.L., Medeiros, B., and Debolskiy, M. A new method for diagnosing effective radiative forcing from aerosol-cloud interactions in climate models (2025). *Atmospheric Chemistry and Physics*. <https://doi.org/10.5194/acp-25-2123-2025>.

SUBMITTED MANUSCRIPTS

2. **Duran, B.M.**, Wall, C.J., Lutsko, N.J., Wynn, S., Watson-Parris, D., and Mülmenstädt, C.J. A new proxy for the LWP adjustment from first principles (preprint, 2026) *ESS Open Archive*. <https://doi.org/10.22541/essoar.15006786/v1>
1. Wynn, S., Watson-Parris, D., **Duran, B.M.**, and Mülmenstädt, C.J. Base-state dependence of parameter sensitivity for the Twomey effect in a perturbed parameter ensemble (preprint, 2026) *Atmospheric Chemistry and Physics*. <https://doi.org/10.5194/egusphere-2026-2811>

AWARDS RECEIVED

2022	National Science Foundation Graduate Research Fellow
2020-2022	National Hispanic Scholar
2021	AGU Student Travel Grant Recipient

CONFERENCE PRESENTATIONS

2026	Invited Talk: Lawrence Livermore National Laboratory (LLNL)
2026	Oral: Scripps Student Symposium (S3)
2026	Oral: CalGFD 2026
2026	Oral: 2026 CESM Workshop
2026	Oral: Weather & Climate Livestream
2026	Oral: CESM Winter Working Group Meeting
2025	Poster: Gordon Research Conference, Radiation & Climate at Bates College
2025	Oral: 2025 CESM Workshop
2025	Invited Talk: Equilibrium Climate Sensitivity and Cloud Feedback Virtual Symposium
2024	Poster: AGU, Washington, D.C.
2024	Poster: Micro2Macro Workshop, University of Wyoming
2024	Oral: CFMIP Conference at Boston University

TEACHING AND MENTORING

Teaching Assistant

2026	<i>Dynamics of Atmos & Climate</i> , UCSD. Instructor: Nicholas Lutsko.
2025	<i>The Atmosphere</i> , UCSD. Instructor: Lynn Russell.
2021	<i>Mechanics and Electromagnetic Phenomena</i> , Georgetown University. Instructor: Patrick Johnson.

Guest Lecturer

2026	<i>Dynamics of Atmos & Climate</i> , UCSD. Instructor: Nicholas Lutsko.
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Research Mentor

Sophie Wynn, PhD, SIO
Devan Velji, undergraduate, UCSD

Peer Mentor

Wenjing Zhang, SIO
David Cera, SIO

SERVICE

Department

2026-	Student Transportation Advisory Committee
2024-	Co-organizer of SIO Climate Journal Club
2024-26	Climate Sciences representative and senior co-chair of the Scripps Graduate Student Council
2024-25	SIO Extreme Weather and Lecturer faculty search committees
2023-24	Co-organizer of the Climate, Atmospheric Sciences and Physical Oceanography (CASPO) seminar series

Other

2024-	Reviewer for <i>Science Advances</i> , <i>JGR: Atmospheres</i> , <i>Nature Climate Change</i> , and <i>Atmospheric Chemistry and Physics</i>
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PROFESSIONAL REFERENCES

Nicholas Lutsko, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: nlutsko@ucsd.edu
Shang-Ping Xie, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: sxie@ucsd.edu
Duncan Watson-Parris, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA: dwatsonparris@ucsd.edu
Sloan Coats, Department of Earth Sciences, University of Hawai'i at Mānoa, Honolulu, HI, USA: scoats@hawaii.edu